Fundamental of Mathematics

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Number System

1. **Natural numbers :** $N = \{1, 2, 3, 4, ...\}$
2. **Whole numbers :** $W = \{0, 1, 2, 3, ...\}$
3. **Integers :** Z or $I = \{... -3, -2, -1, 0, 1, 2, 3, ...\}$
4. **Positive Integers :** Z^+ or $I^+ = \{1, 2, 3, ...\} = N$
5. **Negative Integers :** Z^- or $I^- = \{-1, -2, -3, ...\}$
6. **Non-negative Integers :** $= \{0, 1, 2, 3, ...\} = W$
7. **Non-positive Integers :** $= \{... -3, -2, -1, 0\}$
8. **Even Integers :** $= \{... -6, -4, -2, 0, 2, 4, 6, ...\}$
   $\Rightarrow$ These numbers can be expressed by "$2n$", $n \in I$
9. **Odd Integers :** $= \{... -5, -3, -1, 1, 3, 5, ...\}$
   $\Rightarrow$ These numbers can be expressed by "$2n \pm 1$", $n \in I$

10. **Prime Numbers :**
    Natural numbers which have two distinct factors are prime numbers.

    Ex. 2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, ...

Note:

(i) 1 is not a prime number.
(ii) All prime numbers (except 2, 3) can be written as form "$6n \pm 1$" where $n \in Z$
(iii) 2 is the only prime number which is even.

Point to Remember!!!

(i) 0 is neither positive nor negative integer.

(ii) 0 is an even integer.

1. Composite Numbers :

Natural numbers (except 1) which have more than two distinct factors are composite numbers.

Ex. 4, 6, 8, 10, 12, 14, ...

Point to Remember!!!

(i) 1 is neither prime nor composite.
(ii) 4 is the least composite number.

2. Co-prime (Relative prime) numbers :

Two natural numbers (not necessarily prime) are co-prime if their HCF (GCD) is 1.

Ex. (1, 2), (3, 5), (5, 9), (16, 15), ...

Point to Remember!!!

(i) Two distinct prime numbers are always co-prime but the converse need not be true.
(ii) Consecutive natural numbers are always co-prime.

3. Twin prime numbers :

If the difference between two prime numbers is 2 then they are called twin prime numbers.

Ex. (3, 5), (5, 7), (11, 13), (17, 19), ...

4. Rational numbers :

Numbers which can be written in form of "p/q" where p, q $\in$ Z and q $\neq$ 0. Set of rational numbers is denoted by "Q".

Ex. 2/3, 7/9, −2/5, 7/1, ...

Point to Remember!!!

All terminating decimal numbers, non-terminating but repeating decimal numbers are rational numbers.

5. Irrational numbers :

Real numbers which are not rational, are irrational numbers. The set of irrational numbers is denoted by "Q^c" or " Q' "

Ex. $\sqrt{2}$, $\sqrt{3}$, $\sqrt{5}$,.....π, e,.....

Point to Remember!!!

Non-terminating and non-repeating decimal numbers are irrational numbers

6. Real numbers :

The complete set of rational and irrational numbers is the set of real numbers and it is denoted by "R" hence $R = Q \cup Q^c$

Point to Remember!!!

(i) Square of real number is non-negative.
(ii) Real numbers can be represented on real number line.

17. Complex numbers :

These are the numbers which are represented in the form of a + ib where a, b $\in$ R and $i = \sqrt{-1}$ (iota). Single complex number is denoted by "Z" and Set of all complex numbers is represented by "C".

Ex. 2 + 3i, 5–7i, 0 + 2i, $2-\sqrt{3}i$, 5, ...

Point to Remember!!!

All real numbers are complex numbers.

Q.1 Prove that n^3-n is divisible by 24, where n is odd integer.

Sol. Let n = 2m + 1, m $\in$ I (∵ m is odd integer)

Now $n^3 - n = n(n^2 - 1)$

$= n(n + 1)(n - 1)$ {it is multiple of 3}

$= (2m + 1)(2m + 1 + 1)(2m + 1 - 1)$

$= (2m + 1)\, 2(m + 1)2m$

$= 4m(m + 1)(2m + 1)$

For all m $\in$ I, m(m + 1) (2m +1) will be a multiple of 6.

Hence **$n^3 - n$ is divisible by 24.**

Q.2 Prove that $p^2 - 1$ is divisible by 24, where p is any prime number ≥ 5.

Sol. Let p = 6n ± 1, n $\in$ I (∵ p is a prime number ≥ 5)

hence $p^2 - 1 = (p + 1)(p - 1)$

$= (6n \pm 1 + 1)(6n \pm 1 - 1)$

$= (6n \pm 2)(6n)$

$= 12\, n\, (n \pm 1)$

For all n $\in$ I, n(n ± 1) will be an even number

Hence **$p^2 - 1$ is divisible by 24.**

Q.3 Find a, b $\in$ I given that $a^2 - b^2 = 13$

Sol. $a^2 - b^2 = 13$

$(a + b)(a - b) = 13 = 1 \times 13$ ... (i)

Now let us assume a, b > 0 then from (i)

$$\left.\begin{matrix} a+b=13 \\ a-b=1 \end{matrix}\right\} \Rightarrow a = 7,\ b = 6$$

So, in general a = ± 7, b ± 6

Q.4 Find m, n $\in$ I if $m^2 - n^2 = 2002$.

Sol. ∵ $m^2 - n^2 = 2002$

$(m + n)(m - n) = 2 \times 1001$

⇒ If m, n both even or both odd then (m + n), (m −n) both are even then LHS will be multiple of 4 but RHS is not.
⇒ If m is even and n is odd or vice-versa then (m + n), (m −n) both are odd hence LHS will be odd but RHS is not.
Hence there will be no solution.

Q.5 If $x + y^2 = x^2 + y = 12$, then find (x, y)

Sol. $x^2 + y = 12$ ... (i)

$x + y^2 = 12$ ... (ii)

(i) − (ii) $\Rightarrow (x^2 - y^2) + (y - x) = 0$

$(x - y)\{x + y - 1\} = 0$

$x = y$ or $x = 1 - y$

if $x = y$ then from (i)

$x^2 + x = 12 \Rightarrow x = 3, -4$

hence $x = y = 3, -4$

if $x = 1 - y$ then from (i)

$(1 - y)^2 + y = 12$

$y^2 - y - 11 = 0 \Rightarrow y = \dfrac{1 \pm 3\sqrt{5}}{2}$

$x = \dfrac{1 \mp 3\sqrt{5}}{2}$

$(x, y) \equiv (3, 3), (-4, -4), \left(\dfrac{1 \pm 3\sqrt{5}}{2}, \dfrac{1 \mp 3\sqrt{5}}{2}\right)$

FACTORIZATION

Some useful identities:

1. $a^2 - b^2 = (a + b)(a - b)$
2. $(a \pm b)^2 = a^2 + b^2 \pm 2ab$
3. $a^3 - b^3 = (a - b)(a^2 + ab + b^2) = (a - b)^3 + 3ab(a - b)$

   $a^3 + b^3 = (a + b)(a^2 - ab + b^2) = (a + b)^3 - 3ab(a + b)$
4. $(a - b)^3 = a^3 - b^3 - 3ab(a - b)$

   $(a + b)^3 = a^3 + b^3 + 3ab(a + b)$
5. $(a + b + c)^2 = a^2 + b^2 + c^2 + 2ab + 2bc + 2ca$
6. $(a + b + c + d)^2 = a^2 + b^2 + c^2 + d^2 + 2ab + 2bc + 2cd + 2da + 2ac + 2bd$
7. $a^3 + b^3 + c^3 - 3abc = (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca)$

   → If $a + b + c = 0$, then $a^3 + b^3 + c^3 = 3abc$

8. $a^2+b^2+c^2-ab-bc-ca=\frac{1}{2}\{(a-b)^2+(b-c)^2+(c-a)^2\}$

9. $1+x^2+x^4=(1+x^2)^2-x^2=(1+x+x^2)(1-x+x^2)$

10. $(a+b+c)^3=a^3+b^3+c^3+3(a+b)(b+c)(c+a)$

Q.1 If $a+b=2$ and $a^3+b^3=27$ then find a^2+b^2

Sol. $a^3+b^3=(a+b)^3-3ab(a+b)$

$27=(2)^3-3ab(2)$

$19=-6ab \Rightarrow 2ab=-19/3$

$(a+b)^2=a^2+b^2+2ab$

$2^2=a^2+b^2+(-19/3) \Rightarrow a^2+b^2=$ **31/3**

Q.2 If $a+b+c=4$, $a^2+b^2+c^2=6$ and $a^3+b^3+c^3=8$ then find $a^4+b^4+c^4$

Sol. $(a+b+c)^2=a^2+b^2+c^2+2(ab+bc+ca)$

$16=6+2(\lambda)$

$\Rightarrow \lambda=5$

$a^3+b^3+c^3-3abc=(a+b+c)(a^2+b^2+c^2-\lambda)$

$8-3\mu=(4)(6-5)$

$\Rightarrow \mu=4/3$

$(ab+bc+ca)^2=a^2b^2+b^2c^2+c^2a^2+2abc(a+b+c)$

$$25=\gamma+2\left(\frac{4}{3}\right)(4)$$

$\Rightarrow \gamma=43/3$

$(a^2+b^2+c^2)^2=a^4+b^4+c^4+2(\gamma)$

$36=a^4+b^4+c^4+86/3$

$\Rightarrow a^4+b^4+c^4=$ **22/3**

Q.3 If $x-\frac{1}{x}=3$, then find the value of $2\left(x^3-\frac{1}{x^3}\right)-3\left(x^2+\frac{1}{x^2}\right)-39$

Sol. $\left(x-\frac{1}{x}\right)^2=x^2+\frac{1}{x^2}-2x\frac{1}{x}$

$$9=x^2+\frac{1}{x^2}-2 \Rightarrow x^2+\frac{1}{x^2}=11 \qquad \text{...(i)}$$

$$\left(x^3-\frac{1}{x^3}\right)=\left(x-\frac{1}{x}\right)^3+3x\frac{1}{x}\left(x-\frac{1}{x}\right)$$

$= 27 + 3(3) \Rightarrow x^3 - \frac{1}{x^3} = 36$

Put in required expression $\Rightarrow 2(36) - 3(11) - 39 = \mathbf{0}$

Q.4 Factorize the following expressions:

(i) $x^4 - y^4$ **(ii)** $9a^2 - (2x - y)^2$

Sol. **(i)** $x^4 - y^4 = (x^2)^2 - (y^2)^2$

$= (x^2 + y^2)(x^2 - y^2)$

$= \mathbf{(x^2 + y^2)(x - y)(x + y)}$

(ii) $9a^2 - (2x - y)^2 = (3a)^2 - (2x - y)^2$

$= \mathbf{(3a + 2x - y)(3a - 2x + y)}$

Q.5 Factorize

(i) $8x^3 - 27y^3$ **(ii)** $a^6 - b^6$

Sol. **(i)** $8x^3 - 27y^3 = (2x)^3 - (3y)^3$

$= (2x - 3y)((2x)^2 + (2x)(3y) + (3y)^2)$

$= \mathbf{(2x - 3y)(4x^2 + 6xy + 9y^2)}$

(ii) $a^6 - b^6 = (a^3)^2 - (b^3)^2$

$= (a^3 + b^3)(a^3 - b^3)$

$= \mathbf{(a + b)(a^2 - ab + b^2)(a - b)(a^2 + ab + b^2)}$

Q.6 Factorize

(i) $x^2 + 3x - 40$ **(ii)** $x^2 - 3x - 40$

(iii) $3x^2 - 10x + 8$ **(iv)** $12x^2 + x - 35$

Sol. **(i)** $x^2 + 3x - 40 = x^2 + 8x - 5x - 40$

$= x(x + 8) - 5(x + 8)$

$= \mathbf{(x - 5)(x + 8)}$

(ii) $x^2 - 3x - 40 = x^2 - 8x + 5x - 40$

$= x(x - 8) + 5(x - 8)$

$= \mathbf{(x + 5)(x - 8)}$

(iii) $3x^2 - 10x + 8 = 3x^2 - 6x - 4x + 8$

$= 3x(x - 2) - 4(x - 2)$

$= \mathbf{(3x - 4)(x - 2)}$

(iv) $12x^2 + x - 35 = 12x^2 + 21x - 20x - 35$

$= 3x(4x + 7) - 5(4x + 7)$

$= \mathbf{(3x - 5)(4x + 7)}$

Q.7 Factorize by converting into perfect square:

(i) $a^2 - 4a + 3 + 2b - b^2$

(ii) $a^4 + a^2b^2 + b^4$

Sol. **(i)** $(a^2 - 4a + 4 - 4) + 3 - (b^2 - 2b + 1 - 1)$

$= (a - 2)^2 - 4 + 3 - (b - 1)^2 + 1$

$= (a - 2)^2 - (b - 1)^2$

$= [(a - 2) + (b - 1)]\,[(a - 2) - (b - 1)]$

$= \mathbf{(a + b - 3)\,(a - b - 1)}$

(ii) $a^4 + a^2b^2 + b^4$

$= a^4 + 2a^2b^2 + b^4 - a^2b^2$

$= (a^2 + b^2)^2 - (ab)^2$

$= \mathbf{(a^2 + b^2 + ab)\,(a^2 + b^2 - ab)}$

Q.8 Factorize $8a^3 + b^3 + c^3 - 6abc$

Sol. $(2a)^3 + (b)^3 + (c)^3 - 3(2a)(b)(c)$

$= (2a + b + c)\,((2a)^2 + b^2 + c^2 - (2a)b - bc - c(2a))$

$= \mathbf{(2a + b + c)\,(4a^2 + b^2 + c^2 - 2ab - bc - 2ac)}$

Q.9 Solve the equations :

(i) $4^x - 10.2^{x-1} = 24$ **(ii)** $4 \cdot 2^{2x} - 6^x = 18 \cdot 3^{2x}$

Sol. **(i)** $(2^x)^2 - 10(2^x)\frac{1}{2} = 24$

Let $2^x = t$ then

$\Rightarrow t^2 - 5t - 24 = 0 \Rightarrow t^2 - 8t + 3t - 24 = 0$

$\Rightarrow (t - 8)(t + 3) = 0$

$\Rightarrow t = 8 \Rightarrow 2^x = 2^3 \Rightarrow x = 3$

$\Rightarrow t = -3 \Rightarrow 2^x = -3 \Rightarrow$ Not possible

(ii) $4(2^x)^2 - 2^x \cdot 3^x - 18(3^x)^2 = 0$

Let $2^x = a$, $3^x = b$ then

$\Rightarrow 4a^2 - ab - 18b^2 = 0 \Rightarrow 4a^2 - 9ab + 8ab - 18b^2 = 0$

$\Rightarrow (4a - 9b)(a + 2b) = 0$

$4a = 9b \Rightarrow 4 \cdot 2^x = 9 \cdot 3^x \Rightarrow 2^{x+2} = 3^{x+2} \Rightarrow x + 2 = 0 \Rightarrow x = -2$

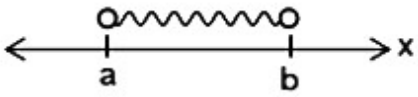
$a = -2b \Rightarrow 2^x = -2 \cdot 3^x \Rightarrow$ Not possible

Intervals

These are basically subset of real numbers. If there are two numbers a, b $\in$ R such that a < b then we can define four types of intervals.

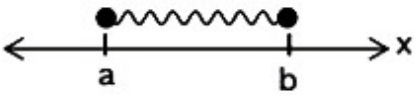
(A) Open Interval :

If a < x < b then it can be written as x $\in$ (a, b).
On number line it can be represented as



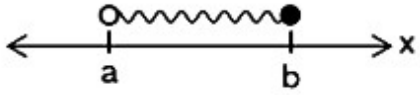
(B) Closed Interval :

If a ≤ x ≤ b then it can be written as x $\in$ [a, b].
On number line it can be represented as

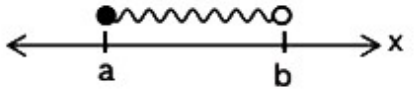


(C) Semi open or Semi closed interval :

If a < x ≤ b then x $\in$ (a, b]

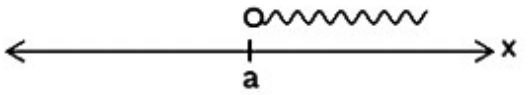


If a ≤ x < b then x $\in$ [a, b)

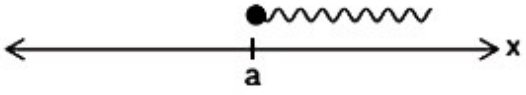


(D) Infinite Intervals :

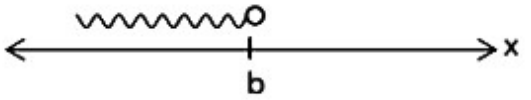
If x > a then x $\in$ (a, ∞)



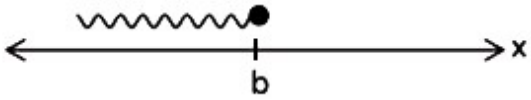
If x ≥ a then x $\in$ [a, ∞)



If x < b then x $\in$ ($-\infty$, b)



If x ≤ b then x $\in$ ($-\infty$, b]



Note:

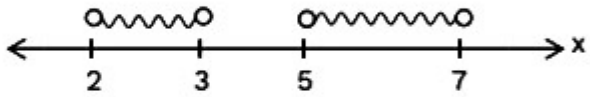
(i) For discrete values of x we use sets

Ex. If x = 1, 2 then we write x $\in$ {1, 2}

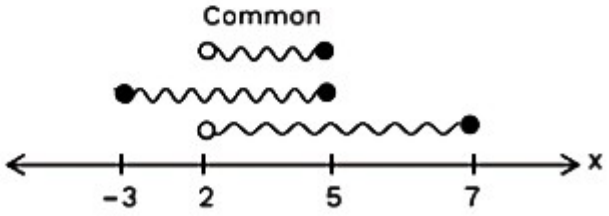
(ii) If there is no value of x then we write x $\in$ ϕ

Operation on Intervals :

1. If $x \in (2, 3)$ or $x \in (5, 7)$ then we write it as $x \in (2, 3) \cup (5, 7)$



2. **Intersection (and) :**
If $x \in (2, 7]$ and $x \in [-3, 5]$ then common values of both intervals come under intersection. Hence intersection will be $x \in (2, 5]$



Remainder Theorem

If polynomial p(x) whose degree is greater than or equal to 1, is divided by (x – a), then the remainder is p(a).

Ex. If $p(x) = x^3 - 7x^2 + 6x + 4$ is divided by $(x - 1)$ then remainder = $p(1) = 1 - 7 + 6 + 4 = 4$

Point to Remember!!!

If p(x) is divided by (ax + b) then remainder is p(-b/a).

Factor Theorem

If polynomial p(x) is of degree ≥ 1 and 'a' be a real constant such that p(a) = 0 then (x – a) will be a factor of p(x).

Ex. Let $p(x) = x^4 - 7x^2 + 6$ is a polynomial and it is such that $p(1) = 0$ then $(x - 1)$ will be a factor of p(x).

Dividend = Quotient × Divisor + Remainder

Q.1 Find the remainder when $p(x) = x^3 + 2x^2 - 3x + 1$ is divided by $2x + 1$.

Sol. By remainder theorem

Remainder = $p(-½)$

$\therefore p(-½) = (-½)^3 + 2(-½)^2 - 3(-½) + 1 = \frac{-1}{8} + \frac{1}{2} + \frac{3}{2} + 1 = \frac{23}{8}$ is remainder.

Q.2 If $p(x) = kx^4 + 2x^2 + 2k + 1$ is divided by $x - 1$ then remainder is $4k - 2$, then find k.

Sol. By remainder theorem

Remainder = $p(1)$

$4k - 2 = k(1)^4 + 2(1)^2 + 2k + 1$

$k = \mathbf{5}$

Q.3 Factorize $p(x) = x^3 - 6x^2 + 11x - 6$

Sol. If $x = 1$ then

$p(1) = (1)^3 - 6(1)^2 + 11(1) - 6$

$= 12 - 12 = 0$

hence $(x - 1)$ is factor of $p(x)$

Similarly $p(2) = 0 \quad \Rightarrow (x - 2)$ is factor of $p(x)$

$p(3) = 0 \quad \Rightarrow (x - 3)$ is factor of $p(x)$

Hence $p(x) = \mathbf{(x - 1)(x - 2)(x - 3)}$

Q.4 If $x^2 + x - 6$ is a factor of $p(x) = 2x^4 + x^3 - ax^2 + bx + a + b - 1$, then find the value of 'a' and 'b'

Sol. $\because x^2 + x - 6 = (x + 3)(x - 2)$

$\therefore (x + 3)$ and $(x - 2)$ both are factors of $p(x)$

$\Rightarrow p(-3) = 0$

$\Rightarrow 2(81) + (-27) - a(9) + b(-3) + a + b - 1 = 0$

$\Rightarrow 8a + 2b = 134$

$\Rightarrow 4a + b = 67$ ...(i)

Also, $p(2) = 0$

$\Rightarrow 2(16) + (8) - a(4) + b(2) + a + b - 1 = 0$

$\Rightarrow 3a - 3b = 39$

$\Rightarrow a - b = 13$ ...(ii)

By (i) and (ii)

$\mathbf{a = 16,} \qquad \mathbf{b = 3}$

Q.5 A polynomial p(x) when divided by x and x – 1 gives remainder 1 and 3 respectively find the remainder when p(x) is divided by $x^2 - x$

Sol. By remainder theorem

$p(0) = 1, \qquad p(1) = 3$

If p(x) is divided by $x^2 - x$ remainder is R(x) = ax + b ($\because$ remainder can have atmost 1-degree as divisor is of degree '2')

Now by division algorithm

Dividend = Quotient × Divisor + Remainder

$p(x) = Q(x) \times x(x-1) + ax + b$ ...(i)

Put x = 1 in (i)

$p(1) = Q(1) \times 0 + a + b$

$3 = a + b$ ...(ii)

Put x = 0 in (i)

$p(0) = Q(0) \times 0 + b$

$1 = b$ ...(iii)

By (ii) and (iii)

$\Rightarrow$ $a = 2, \qquad b = 1$

$\therefore$ Remainder = **$2x + 1$**

Q.6 Factorize $(x + 1)(x + 2)(x + 3)(x + 4) - 8$

Sol. $\{(x + 1)(x + 4)\}\{(x + 2)(x + 3)\} - 8$

$= (x^2 + 5x + 4)(x^2 + 5x + 6) - 8$

Let $x^2 + 5x = t$ then

$\Rightarrow (t + 4)(t + 6) - 8$

$= (t^2 + 10t + 16)$

$= (t + 2)(t + 8)$

$=$ **$(x^2 + 5x + 2)(x^2 + 5x + 8)$**

Q.7 Factorize $f(x) = x^3 + 3x^2 + 3x + 2$

Sol. Put $x = -2$

$f(-2) = (-8) + 3(4) + 3(-2) + 2 = 0$

$\because$ $f(-2) = 0 \Rightarrow (x + 2)$ is a factor of f(x)

Now,
$$
\begin{array}{r}
x^2 + x + 1 \\
(x+2)\overline{)\,x^3 + 3x^2 + 3x + 2} \\
\underline{x^3 + 2x^2} \\
x^2 + 3x + 2 \\
\underline{x^2 + 2x} \\
x + 2 \\
\underline{x + 2} \\
\times \quad \times
\end{array}
$$

$\therefore$ $(x^3 + 3x^2 + 3x + 2) =$ **$(x + 2)(x^2 + x + 1)$**

Divisibility Rules

For 2: Last digit of number must be 0, 2, 4, 6 or 8

For 3: Sum of all digits of the number must be a multiple of 3.

For 4: Number formed by last two digits must be a multiple of 4.

For 5: Unit digit of number must be 0 or 5.

For 6: Number must be divisible by both 2 and 3.

For 8: Number formed by last three digits of number must be a multiple of 8.

For 9: Sum of all digits of the number must be a multiple of 9.

For 10: Last digit of number must be 0.

For 11: (Sum of digits at even places) – (Sum of digits at odd places) = Multiple of 11.

Q.1 If the number 5279ab is divisible by 18 then (a, b) is equal to

Sol. The number must be divisible by 2 and 9 both

For 2 $\Rightarrow$ Digit b must be 0, 2, 4, 6 or 8

For 9 $\Rightarrow$ $5 + 2 + 7 + 9 + a + b = 9k,\ k \in w$

$23 + a + b = 9k$

if $b = 0$ then $a = 4$

if $b = 2$ then $a = 2$

$\vdots$

$(a, b) \equiv (4, 0), (2, 2), (0, 4), (9, 4), (7, 6), (5, 8)$

RATIO

It is a comparison of two quantities by division.

If a and b are two numbers then ratio of a and b is $\frac{a}{b}$ or $a \div b$ and it is denoted by $a : b$

Where a $\Rightarrow$ antecedent

b $\Rightarrow$ consequent

Ex. $\frac{3}{5}, \frac{7}{9}, \frac{2}{7}, \ldots$

PROPORTION

The equality of two ratios is called proportion If $\frac{a}{b} = \frac{c}{d}$ then a, b, c and d are said to be in proportion and we write a : b : : c : d and each term $\frac{a}{b}, \frac{c}{d}$ is called proportional. Also a, b, c, d are called first, second, third and fourth proportional respectively.

Ex. $\frac{3}{4} = \frac{6}{8}$ we write it 3 : 4 : : 6 : 8

Note : If $\frac{a}{b} = \frac{c}{d}$ then

(i) $\frac{a+b}{b} = \frac{c+d}{d}$ (Componendo)

(ii) $\frac{a-b}{b} = \frac{c-d}{d}$ (Dividendo)

(iii) $\frac{a+b}{a-b} = \frac{c+d}{c-d}$ (Componendo and Dividendo)

Note :

If $\frac{a}{b} = \frac{c}{d} = \frac{e}{f} = \lambda$ (let) then,

$\lambda = \frac{a+c+e}{b+d+f}$

Note :

If a : b : : c : d then
Product of means = Product of extremes
i.e., b·c = a·d

Note : (i) Fourth Proportional:
if x is fourth proportional of a, b, c then a : b : : c : x $\Rightarrow$ $x = \frac{bc}{a}$

(ii) Third Proportional:
if x is third proportional of a, b then a : b : : b : x $\Rightarrow$ $x = \frac{b^2}{a}$

(iii) Mean Proportional:
If x is mean (or second) proportional of a, b the
a : x : : x : b $\Rightarrow$ $x = \sqrt{ab}$

Q.1 If $x : y = 1 : 2$ then find the value of $(2x + 3y) : (x + y)$

Sol. $\because \frac{x}{y} = \frac{1}{2} \quad \Rightarrow \quad$ Let $x = k, \quad y = 2k \quad (k \neq 0)$

$$\therefore \frac{2x + 3y}{x + y} = \frac{2(k) + 3(2k)}{k + 2k} \Rightarrow \frac{8k}{3k} = \frac{8}{3}$$

$\therefore$ required ratio is 8 : 3.

Q.2 If $15(2x^2 - y^2) = 7xy$ then find the value of $x : y$.

Sol. Let $\frac{x}{y} = k \Rightarrow x = ky$

Now from given equation

$15\{2(ky)^2 - y^2\} = 7\,(ky)y$

$\Rightarrow 15y^2\{2k^2 - 1\} = 7ky^2$

$\Rightarrow 15(2k^2 - 1) = 7\,k$

$\Rightarrow 30k^2 - 7k - 15 = 0$

$\Rightarrow 30k^2 - 25k + 18k - 15 = 0$

$\Rightarrow 5k\,(6k - 5) + 3(6k - 5) = 0$

$\Rightarrow (5k + 3)\,(6k - 5) = 0$

$\Rightarrow k = -\frac{3}{5} \quad$ or $\quad k = \frac{5}{6}$

$\therefore \frac{x}{y} = -\frac{3}{5} \quad$ or $\quad \frac{5}{6}$

Q.3 If $\frac{a}{b} = \frac{c}{d} = \frac{e}{f} = \lambda$ (all numbers are positive)

Then prove that (i) $\lambda = \sqrt{\frac{ac}{bd}} \qquad$ (ii) $\lambda = \left\{\frac{a^2 + c^2 + e^2}{b^2 + d^2 + f^2}\right\}^{1/2}$

Sol. $a = b\lambda, \quad c = d\lambda, \quad e = f\lambda$

(i) $\sqrt{\frac{ac}{bd}} = \sqrt{\frac{(b\lambda)(d\lambda)}{bd}} = \sqrt{\lambda^2} = \lambda$

(ii) $a^2 = b^2\lambda^2 \quad \ldots(1)$

$c^2 = d^2\lambda^2 \quad \ldots(2)$

$e^2 = f^2\lambda^2 \quad \ldots(3)$

Add all three equations, $a^2 + c^2 + e^2 = \left(b^2 + d^2 + f^2\right)\lambda^2$

$$\Rightarrow \lambda = \left\{\frac{a^2 + c^2 + e^2}{b^2 + d^2 + f^2}\right\}^{1/2}$$

INDICES

Some useful formulae :

1. $a^m \cdot a^n = a^{m+n}$
2. $a^m \div a^n = a^{m-n}$
3. $(a^m)^n = a^{mn}$
4. $\left(\frac{a}{b}\right)^{-m/n} = \left(\frac{b}{a}\right)^{m/n}$
5. $a^m \div b^{-n} = a^m \cdot b^n$
6. $a^0 = 1 \quad (a \neq 0)$

Q. Evaluate (i) $\left(\sqrt[6]{64}\right)^{-1/2}$ (ii) $\left(\frac{121}{169}\right)^{-3/2}$

Sol. (i) $\because 64 = 2^6 \Rightarrow \sqrt[6]{64} = 2$

$$\therefore \text{ given exp.} = (2)^{-1/2} = \left(\frac{1}{2}\right)^{1/2} = \frac{1}{\sqrt{2}}$$

(ii) $$\left(\frac{121}{169}\right)^{-3/2} = \left(\frac{11^2}{13^2}\right)^{-3/2} = \left(\frac{11}{13}\right)^{2\left(-\frac{3}{2}\right)} = \frac{13^3}{11^3}$$